CBSE Class 12 Physics — Half-Yearly Predicted Paper (Set 5 of 10)

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQ/Assertion-Reason/VSA)

Q1. The electrostatic force between two point charges placed at a distance r in a vacuum is F. If the distance between them is strictly halved while keeping the charges unchanged, the new electrostatic force will be: (a) $F/2$ (b) $F/4$ (c) $2F$ (d) $4F$ [PYQ 2021] Very High

Q2. For a uniform electric field strictly directed along the positive x-axis, the corresponding equipotential surfaces are: (a) planes parallel to the x-y plane. (b) planes parallel to the y-z plane. (c) planes parallel to the x-z plane. (d) concentric spheres centered at the origin. [PYQ 2020] Very High

Q3. The drift velocity (v_d) of free electrons in a conducting wire of length L subjected to a constant potential difference V is directly proportional to: (a) L (b) L^2 (c) $1/L$ (d) $1/L^2$ [PYQ 2022] High

Q4. A circular coil of N turns and radius R carries a steady current I. If the radius of the coil is perfectly halved and the number of turns is doubled while passing the exact same current, the new magnetic field at its centre will become: (a) 2 times (b) 4 times (c) 8 times (d) Unchanged [Practice] Very High

Q5. To convert a moving coil galvanometer into a voltmeter of a given range, one must physically connect a: (a) low resistance strictly in series. (b) high resistance strictly in parallel. (c) high resistance strictly in series. (d) low resistance strictly in parallel. [PYQ 2020] Very High

Q6. The correct dimensional formula for magnetic flux (Φ_B) is: (a) $[ML^2T^{-2}A^{-1}]$ (b) $[MLT^{-2}A^{-1}]$ (c) $[ML^2T^{-1}A^{-2}]$ (d) $[ML^2T^{-2}A^{-2}]$ [Practice] Medium

Q7. In an AC circuit, the instantaneous voltage and current are given by $V = 100\sin(100t)$ volts and $I = 10\sin(100t + \pi/3)$ amperes. The average power dissipated in the circuit is: (a) 1000 W (b) 500 W (c) 250 W (d) Zero [PYQ 2019] High

Q8. In a plane electromagnetic wave propagating in a vacuum, the ratio of the average energy density of the electric field to that of the magnetic field is exactly: (a) $1:1$ (b) $c:1$ (c) $1:c$ (d) $c^2:1$ [PYQ 2023] Very High

Q9. A thin glass convex lens of refractive index 1.5 has a focal length f in air. If it is completely immersed in a transparent liquid of the exact same refractive index (1.5), its new focal length will become: (a) $f/2$ (b) $2f$ (c) Zero (d) Infinity [PYQ 2022] Very High

Q10. In Young's double-slit experiment, the condition for strictly constructive interference at a point on the screen is that the phase difference ($\Delta\phi$) between the two interfering waves must be: (a) an odd multiple of π (b) an even multiple of π (c) an odd multiple of $\pi/2$ (d) an even multiple of $\pi/2$ [PYQ 2018] Very High

Q11. The stopping potential in a photoelectric effect experiment fundamentally depends upon: (a) only the intensity of incident light. (b) only the frequency of incident light. (c) both the frequency of incident light and the nature of the material. (d) the time of irradiation of light. [PYQ 2020] Very High

Q12. A photon and a free electron have the exact same kinetic energy E. The ratio of the de-Broglie wavelength of the photon to that of the electron is directly proportional to: (a) $\sqrt{E}$ (b) $1/\sqrt{E}$ (c) E (d) $1/E$ [PYQ 2019] High

Q13. Assertion (A): The magnitude of the electrostatic force experienced by a proton and an electron placed in the exact same uniform electric field is identical. **Reason (R):** Both a proton and an electron possess the exact same magnitude of elementary electric charge e. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is also false. [Practice] Very High

Q14. Assertion (A): Bending a uniform conducting wire into a circular loop strictly does not change its macroscopic electrical resistance. **Reason (R):** Electrical resistance depends entirely on the resistivity of the material, its total length, and its cross-sectional area, which remain completely unaltered upon bending. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is also false. [Practice] High

Q15. Assertion (A): An ideal inductor completely blocks steady direct current (DC) but allows high-frequency alternating current (AC) to pass easily. **Reason (R):** The inductive reactance of a coil is given by $X_L = \omega L$, which evaluates to zero for DC and increases strictly proportionally with the frequency of AC. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is true. [PYQ 2022] Very High

Q16. Assertion (A): A diamond sparkles brilliantly largely because of the phenomenon of total internal reflection. **Reason (R):** Diamond has a considerably low absolute refractive index compared to glass, making its critical angle very large. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is also false. [PYQ 2017] High

Q17. What is the exact value of the electric field intensity deep inside a solid, highly charged metallic conductor? [PYQ 2020] Very High

Q18. In which specific direction is the induced magnetic dipole moment of a diamagnetic substance aligned relative to the applied external magnetic field? [PYQ 2018] High

Q19. Write the mathematical formula for the speed of an electromagnetic wave (v) in a material medium in terms of its absolute permeability (μ) and absolute permittivity (ϵ). [PYQ 2019] Very High

Q20. What is the defined rest mass of a photon? [Practice] High

2-Mark Questions

Q21. An electric dipole of length 1 cm and possessing charges $\pm 2 \ \mu\text{C}$ is placed at an angle of 30° with respect to a uniform electric field of 10^5 N/C. Calculate the exact magnitude of the torque experienced by the dipole. [PYQ 2020] Very High

Q22. A parallel plate capacitor has a capacitance of $5 \ \mu\text{F}$ when air is present between its plates. When the entire space is completely filled with a specific liquid, its capacitance increases to $30 \ \mu\text{F}$. Calculate the exact dielectric constant of the liquid. [PYQ 2015] High

Q23. Calculate the exact magnitude of the magnetic field at the centre of a semi-circular current loop of radius 10 cm carrying a steady current of 2 A. (Take $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$). [PYQ 2017] High

Q24. State the Ampere-Maxwell law mathematically. Briefly explain the physical necessity that led James Clerk Maxwell to introduce the concept of 'displacement current'. [PYQ 2018] Very High

Q25. Determine the exact de-Broglie wavelength inherently associated with an electron accelerated from rest through a potential difference of exactly 100 V. [PYQ 2022] Very High

Q26. When a monochromatic ray of light travels from an optically denser medium into a rarer medium, does its characteristic colour change? Give a strict scientific reason for your answer. [PYQ 2019] High

Q27. Give two solid physical reasons why two independent electric field lines can absolutely never cross (intersect) each other in space. [PYQ 2014] Very High

Q28. State the precise mathematical condition for electrical resonance in a series LCR circuit. Calculate the resonant frequency if $L = 100 \text{ mH}$ and $C = 10 \ \mu\text{F}$. [Practice] High

Q29. Calculate the exact linear momentum of a photon of green light having a wavelength of 500 nm. (Take $h = 6.63 \times 10^{-34} \text{ J s}$). [PYQ 2016] Medium

Q30. Two incandescent light bulbs rated at $60 \text{ W}, 220 \text{ V}$ and $100 \text{ W}, 220 \text{ V}$ are connected strictly in series across a 220 V supply. Which of the two bulbs will visibly glow brighter and why? [PYQ 2015] High

Q31. Define the 'current sensitivity' of a moving coil galvanometer. State its SI unit. [PYQ 2020] Very High

Q32. Define 'displacement current'. How does it fundamentally differ from conduction current? [PYQ 2019] Very High

3-Mark Questions

Q33. A point charge $+Q$ is placed exactly at the centre of a spherical Gaussian surface of radius R. The electric flux through the surface is measured to be Φ. Determine the new electric flux if: (i) the radius of the spherical surface is explicitly doubled to $2R$. (ii) the enclosed charge is halved to $+Q/2$. (iii) the spherical surface is entirely replaced by a cubical Gaussian surface of side R enclosing the same charge $+Q$ at its centre. [PYQ 2018] Very High

Q34. A nichrome heating element connected to a $230\ \mathrm{V}$ DC supply draws an initial current of $3.2\ \mathrm{A}$ which quickly settles after a few seconds to a steady value of $2.8\ \mathrm{A}$. Calculate the steady temperature of the heating element. (Given: Room temperature is $27^\circ\mathrm{C}$ and the temperature coefficient of resistance of nichrome averaged over the temperature range involved is Double exponent: use braces to clarify). [PYQ 2020] High

Q35. Define the current sensitivity (I_s) and voltage sensitivity (V_s) of a moving coil galvanometer. If the number of turns of the galvanometer coil is strictly doubled (keeping the area and magnetic field constant), explain mathematically what effect this will have on its current sensitivity and voltage sensitivity. [PYQ 2022] Very High

Q36. Define the term 'self-inductance' of a coil. Show mathematically that the magnetic potential energy stored in an inductor of self-inductance L carrying a steady current I is strictly given by $U = \frac{1}{2}LI^2$. [PYQ 2017] Very High

Q37. What is meant by the phenomenon of 'Total Internal Reflection' in optics? State the two essential conditions required for this phenomenon to actively occur. Derive the exact mathematical relation between the critical angle (i_c) and the refractive index (μ) of the denser medium. [PYQ 2019] Very High

Q38. In Young's double-slit experiment, the two identical slits are artificially modified such that their widths are in the strict ratio of $1 : 4$. Assuming that the intensity of light transmitted is directly proportional to the slit width, calculate the exact ratio of the maximum intensity to the minimum intensity ($I_{max} : I_{min}$) in the resulting interference pattern. [PYQ 2016] High

Q39. A photon and a moving electron possess the exact same de-Broglie wavelength. Which of the two fundamental particles possesses greater kinetic energy? Justify your answer with a rigorous mathematical proof. [PYQ 2020] Very High

Q40. Use Kirchhoff's rules to calculate the exact current flowing through the $10\ \Omega$ resistor in a circuit where a $5\ \mathrm{V}$ battery (internal resistance $1\ \Omega$) and a $2\ \mathrm{V}$ battery (internal resistance $2\ \Omega$) are connected in parallel, with their positive terminals joined together, supplying current to the external $10\ \Omega$ resistor. [PYQ 2015] High

Q41. State any three distinct physical properties of electromagnetic waves. [PYQ 2014] Very High

Q42. Define the 'drift velocity' of free electrons. Derive the exact mathematical relationship between the macroscopic current density (J) and the drift velocity (v_d) in a metallic conductor. [PYQ 2022] Very High

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5-Mark Questions (Long Answer / Case-Study)

Q43. Case Study: Cells in Series and Parallel When multiple electrochemical cells are connected in series, their equivalent emf heavily increases, providing a higher net voltage to an external circuit. Conversely, when connected in parallel, the equivalent emf remains identically the same as a single cell, but the effective internal resistance sharply decreases, allowing a higher current to be drawn without dropping the terminal voltage significantly. Real cells possess an internal resistance r. When a cell delivers a current I, the terminal voltage V is strictly less than the emf E. (A) Write the precise mathematical condition for drawing maximum power/current from a combination of cells having an equivalent internal resistance r_{eq} across an external load R. (1 Mark) (B) Two identical cells, each of emf $1.5\ \mathrm{V}$ and internal resistance $0.5\ \Omega$, are connected strictly in parallel across an external circuit. What is the equivalent emf of this parallel combination? (1 Mark) (C) A single cell of emf E and internal resistance r is connected across a variable external resistor R. Plot a neatly labelled graph showing the variation of terminal voltage V with R. What does the terminal voltage V approach when $R \to \infty$? (2 Marks) [Practice / PYQ 2019] | **[Chapter 3: Current Electricity | Confidence: High]**

Q44. Case Study: Transformers in Power Transmission A transformer is an electrical device fundamentally used to increase (step-up) or decrease (step-down) the alternating voltage in an AC circuit. It operates solely on the principle of mutual induction. The core of a transformer is laminated to significantly minimize energy losses. Largescale transmission of electrical energy over long distances is done at extremely high voltages to minimize I^2R power losses in the transmission cables. (A) Can a standard transformer be used to step-up a steady DC voltage? Give a direct reason. (1 Mark) (B) Give the physical reason why the soft iron core of a transformer is specifically laminated. (1 Mark) (C) An ideal step-down transformer has a primary voltage of $220\ \mathrm{V}$ and steps it down to $22\ \mathrm{V}$. If the primary coil consists of 500 turns, calculate the exact number of turns in the secondary coil and the transformation ratio (k). (2 Marks) [PYQ 2022] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q45. (A) Define the physical term 'capacitance'. Write its SI unit. (B) A parallel plate capacitor of capacitance C_0 is fully charged by a battery to a potential V_0. The battery remains completely connected to the capacitor. A solid dielectric slab of dielectric constant K ($K > 1$) is now slowly and completely inserted between the plates. Derive or state exactly how the following physical quantities will change: (i) Potential difference between the plates. (ii) Capacitance of the capacitor. (iii) Charge stored on the plates. (iv) Electric field between the plates. (v) Electrostatic energy stored in the capacitor. [PYQ 2018, 2020] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q46. (A) State the underlying principle of an AC generator. (B) Draw a neatly labelled schematic diagram of a typical AC generator. (C) Using Faraday's laws of electromagnetic induction, derive the mathematical expression for the instantaneous induced emf produced in the rotating coil. (D) Plot a graph showing the explicit variation of this induced emf with time for one complete revolution of the armature coil. [PYQ 2015, 2019] | **[Chapter 6: Electromagnetic Induction / Ch 7 | Confidence: Very High]**

Q47. (A) Draw a neatly labelled ray diagram of a compound microscope forming its final highly magnified image exactly at the least distance of distinct vision (D). (B) Using the ray diagram, derive the mathematical expression for its total magnifying power. (C) Why are both the objective lens and the eyepiece of a compound microscope strictly chosen to have very short focal lengths? Explain physically. [PYQ 2017, 2023] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Q48. (A) State the Biot-Savart law expressing the magnetic field due to a current element in vector form. (B) Using the Biot-Savart law, systematically derive the mathematical expression for the magnetic field intensity at a point on the axis of a circular coil of radius R carrying a steady current I, situated at a distance x from its centre. (C) Show mathematically that for $x \gg R$, the circular coil completely behaves as a magnetic dipole. [PYQ 2016, 2022] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q49. (A) State Huygens' principle of secondary wavelets. (B) A plane wavefront is incident at an angle i on a plane surface separating an optically rarer medium (speed v_1) from an optically denser medium (speed v_2). Using Huygens' geometrical construction, draw the refracted wavefront. (C) Use this diagram to rigorously verify Snell's law of refraction ($\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\mu_2}{\mu_1}$). [PYQ 2015, 2020] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Q50. (A) Write down Einstein's photoelectric equation explaining clearly all the physical symbols used. (B) Draw a standard graph showing the variation of the maximum kinetic energy of emitted photoelectrons (K_{max}) versus the frequency (ν) of the incident light for a given photosensitive material. (C) Explain exactly how the value of Planck's constant (h) and the work function (Φ) of the material can be extracted from this graph. (D) Does the threshold frequency depend on the intensity of incident light? Justify briefly. [PYQ 2018, 2023] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Answer Key

Q1: (d) $4F$. Electrostatic force $F \propto 1/r^2$. If distance r becomes $r/2$, the new force $F' \propto 1/(r/2)^2 = 4/r^2 = 4F$. **Q2:** (b) planes parallel to the y-z plane. Equipotential surfaces are always perfectly perpendicular to the direction of the uniform electric field. **Q3:** (c) $1/L$. Drift velocity $v_d = \frac{eV\tau}{mL}$. Since voltage V is constant, v_d is inversely proportional to length L. **Q4:** (b) 4 times. Magnetic field for N turns is $B = \frac{\mu_0 NI}{2R}$. New radius $R' = R/2$, new turns $N' = 2N$. $B' = \frac{\mu_0(2N)I}{2(R/2)} = 4 \times \frac{\mu_0 NI}{2R} = 4B$. **Q5:** (c) high resistance strictly in series. This ensures minimum current is drawn from the main circuit so the voltage reading remains accurate. **Q6:** (a) $[ML^2T^{-2}A^{-1}]$. $\Phi_B = B \cdot A = (F/IL) \cdot A = ([MLT^{-2}]/[AL]) \times [L^2] = [ML^2T^{-2}A^{-1}]$. **Q7:** (c) $250\ \mathrm{W}$. $V_{rms} = 100/\sqrt{2}$, $I_{rms} = 10/\sqrt{2}$, Phase diff $\phi = 60°$. Power $P = V_{rms}I_{rms}\cos 60° = \frac{100}{\sqrt{2}} \times \frac{10}{\sqrt{2}} \times \frac{1}{2} = \frac{1000}{4} = 250\ \mathrm{W}$. **Q8:** (a)

$1:1$. In any electromagnetic wave, the total average energy density is shared exactly equally between the electric and magnetic fields. **Q9:** (d) Infinity. By Lens Maker's formula: $\frac{1}{f} = \left(\frac{\mu_g}{\mu_l} - 1\right)(\ldots)$. Since $\mu_g = \mu_l = 1.5$, $\frac{1}{f} = (1-1)(\ldots) = 0 \implies f = \infty$ (it acts as a plane glass plate). **Q10:** (b) an even multiple of π. Constructive interference requires phase difference $\Delta\phi = 2n\pi$, where $n = 0, 1, 2\ldots$ **Q11:** (c) both the frequency of incident light and the nature of the material. ($eV_0 = h\nu - \Phi$). **Q12:** (b) $1/\sqrt{E}$. $\lambda_p = hc/E$. $\lambda_e = h/\sqrt{2mE}$. Ratio $\frac{\lambda_p}{\lambda_e} = \frac{hc/E}{h/\sqrt{2mE}} = c\sqrt{\frac{2m}{E}}$. Thus, Ratio $\propto 1/\sqrt{E}$. **Q13:** (a) Both A and R are true and R is the correct explanation of A. Force magnitude $F = |q|E$. Since $|q| = e$ for both, forces are exactly equal. **Q14:** (a) Both A and R are true and R is the correct explanation of A. Bending does not stretch or compress the cross-section/length. **Q15:** (d) A is false and R is true. An inductor BLOCKS AC (high X_L) and ALLOWS DC ($X_L = 0$). Assertion stated the opposite, making it false. **Q16:** (c) A is true but R is false. Diamond sparkles due to TIR, but it has a very *high* refractive index (~ 2.42), resulting in a *small* critical angle ($\sim 24^\circ$).

Q17: Exactly zero. Charges strictly reside on the outer surface of a conductor in electrostatic equilibrium. **Q18:** Strictly in the opposite direction to the applied external magnetic field. **Q19:** $v = \frac{1}{\sqrt{\mu\epsilon}}$. **Q20:** Strictly zero. A photon can never be at rest.

Q21: $p = q \times 2a = (2 \times 10^{-6}) \times (0.01) = 2 \times 10^{-8}\ \mathrm{C\,m}$. Torque $\tau = pE\sin\theta = (2 \times 10^{-8}) \times (10^5) \times \sin(30^\circ) = 2 \times 10^{-3} \times 0.5 = 10^{-3}\ \mathrm{N\,m}$. **Q22:** Dielectric constant $K = \frac{C_m}{C_0} = \frac{30\ \mu\mathrm{F}}{5\ \mu\mathrm{F}} = 6$. **Q23:** $B = \frac{\mu_0 I}{4R} = \frac{4\pi \times 10^{-7} \times 2}{4 \times 0.10} = \frac{8\pi \times 10^{-7}}{0.4} = 2\pi \times 10^{-6}\ \mathrm{T}$. **Q24:** $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_c + \mu_0\epsilon_0 \frac{d\Phi_E}{dt}$. Maxwell found a logical inconsistency in Ampere's law when calculating the magnetic field near a charging capacitor; a changing electric flux must act as a 'displacement current' to maintain continuity of current across the gap. **Q25:** $\lambda = \frac{1.227}{\sqrt{V}}\ \mathrm{nm} = \frac{1.227}{\sqrt{100}} = \frac{1.227}{10} = 0.1227\ \mathrm{nm}$ (or 1.227 \AA). **Q26:** No. Colour is determined solely by the *frequency* of light, which is an inherent property of the source and strictly remains completely unchanged during refraction across any media. **Q27:** If they crossed, a tangent at the point of intersection would give two different directions for the net electric field at a single point, which is physically impossible. **Q28:** Condition: $X_L = X_C \implies \omega L = 1/\omega C$. Resonant frequency $f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.1 \times 10 \times 10^{-6}}} = \frac{1}{2\pi\sqrt{10^{-6}}} = \frac{1000}{2\pi} \approx 159\ \mathrm{Hz}$. **Q29:** Momentum $p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{500 \times 10^{-9}} = \frac{6.63 \times 10^{-34}}{5 \times 10^{-7}} = 1.326 \times 10^{-27}\ \mathrm{kg\,m/s}$. **Q30:** The $60\ \mathrm{W}$ bulb glows brighter. Resistance $R \propto 1/P$, so $R_{60} > R_{100}$. In series, current I is same. Power dissipated $P' = I^2R$. Since $R_{60} > R_{100}$, the $60\mathrm{W}$ bulb dissipates more power and glows visibly brighter. **Q31:** The deflection produced in the galvanometer per unit current flowing through it. $I_s = \theta/I = NAB/k$. SI Unit: $\mathrm{rad/A}$ (or $\mathrm{divisions/A}$). **Q32:** Displacement current is the equivalent current produced due to the time-varying electric field (or electric flux) in a region of space. Conduction current is due to the actual physical flow of electrons; displacement current exists purely due to changing electric flux in a vacuum/dielectric.

Q33: (i) Remains exactly Φ (Flux is completely independent of the size/shape of the surface). (ii) Becomes $\Phi/2$ (Flux is directly proportional to enclosed charge). (iii) Remains exactly Φ (Total flux depends strictly on enclosed charge, not on geometry). **Q34:** $R_{initial} = V/I_i = 230/3.2 = 71.875\ \Omega$ at $T_0 = 27^\circ\mathrm{C}$. $R_{final} = V/I_f = 230/2.8 \approx 82.14\ \Omega$ at steady temp T. Using $R_f = R_i[1 + \alpha(T - T_0)] \implies 82.14 = 71.875[1 + 1.7 \times 10^{-4}(T - 27)]$. $\frac{82.14}{71.875} - 1 = 1.7 \times 10^{-4}(T - 27) \implies 0.1428 = 1.7 \times 10^{-4}(T - 27) \implies T - 27 \approx 840 \implies T \approx 867^\circ\mathrm{C}$. **Q35:** $I_s = \frac{NAB}{k}$, $V_s = \frac{I_s}{R} = \frac{NAB}{kR}$. If $N \to 2N$, Current Sensitivity heavily doubles ($2I_s$). However, doubling turns requires doubling the length of wire, which precisely doubles the resistance ($R \to 2R$). Thus, Voltage Sensitivity $V_s' = \frac{(2N)AB}{k(2R)} = \frac{NAB}{kR} = V_s$ (remains entirely unchanged). **Q36:** The property by virtue of which a coil opposes any change in the current flowing through it. Induced emf $e = -L\frac{dI}{dt}$. Work done against this emf in time dt is $dW = |e|Idt = (L\frac{dI}{dt})Idt = LIdI$. Integrating from 0 to I: $U = \int_0^I LIdI = \frac{1}{2}LI^2$. **Q37:** When light travels from a denser to a rarer medium and strikes the interface at an angle strictly greater than the critical angle, it is entirely reflected back into the denser medium. Conditions: 1. Light must travel from Denser to Rarer medium. 2. Angle of incidence $i > i_c$. Derivation: By Snell's Law at critical angle, $\mu_d \sin i_c = \mu_r \sin 90^\circ$. Thus $\sin i_c = \mu_r/\mu_d$. If rarer is air ($\mu_r = 1$), $\mu_d = 1/\sin i_c$. **Q38:** Intensity $I \propto \mathrm{width}(W)$. Thus $I_1/I_2 = 1/4$. Amplitude $a \propto \sqrt{I}$. Thus $a_1/a_2 = \sqrt{1/4} = 1/2$. Let $a_1 = a$, $a_2 = 2a$. $I_{max} \propto (a_1 + a_2)^2 = (3a)^2 = 9a^2$.

$I_{min} \propto (a_2 - a_1)^2 = (2a - a)^2 = a^2$. Ratio $I_{max} : I_{min} = 9 : 1$. **Q39:** Let $\lambda_p = \lambda_e = \lambda$. Momentum $p = h/\lambda$. Therefore, both strictly have identical momentum p. Kinetic energy $K = p^2/2m$. Since p is identical and mass of electron (m_e) is vastly smaller than mass of photon (which is effectively equivalent to E/c^2, or for particles $m_e \ll m_{proton}$), wait, photon has strictly zero rest mass. Energy of photon $E_p = pc$. Energy of electron $K_e = p^2/2m$. Ratio $E_p/K_e = (pc)/(p^2/2m) = 2mc/p = 2mc\lambda/h$. For normal energetic electrons, $E_p \gg K_e$. The photon possesses vastly greater energy. **Q40:** Use Kirchhoff's loop rules. Left loop (5V battery, 1Ω, and 10Ω): $5 - 1 \cdot I_1 - 10(I_1 + I_2) = 0 \implies 11I_1 + 10I_2 = 5$. Right loop (2V battery, 2Ω, and 10Ω): $2 - 2 \cdot I_2 - 10(I_1 + I_2) = 0 \implies 10I_1 + 12I_2 = 2 \implies 5I_1 + 6I_2 = 1$. Solving: multiply second by $2.2 \implies 11I_1 + 13.2I_2 = 2.2$. Subtracting first: $3.2I_2 = -2.8 \implies I_2 = -0.875\ \text{A}$. Thus $11I_1 + 10(-0.875) = 5 \implies 11I_1 = 13.75 \implies I_1 = 1.25\ \text{A}$. Current through $10\Omega = I_1 + I_2 = 1.25 - 0.875 = 0.375\ \text{A}$. **Q41:** 1. They are completely transverse in nature. 2. They do not strictly require any material medium to propagate (travel at c in vacuum). 3. They transport both energy and momentum through space. **Q42:** The average uniform velocity with which free electrons drift towards the positive terminal of a conductor under an applied electric field. Current $I = neAv_d$. Current density $J = I/A = (neAv_d)/A = nev_d$.

Q43: (A) Maximum current is drawn when External Load Resistance (R) = Equivalent Internal Resistance (r_{eq}). (B) $1.5\ \text{V}$. (C) Plot V on y-axis, R on x-axis. $V = E - Ir = E - (E/(R + r))r = E(\frac{R}{R+r})$. The curve starts from origin ($R = 0, V = 0$), sharply rises, and asymptotically approaches a strict maximum horizontal line $V = E$ as $R \to \infty$. **Q44:** (A) No, because DC provides a constant magnetic flux. Mutual induction heavily strictly requires a continuously *changing* magnetic flux. (B) To increase electrical resistance within the core cross-section, which strictly minimizes power dissipated as heat due to induced eddy currents. (C) Transformation ratio $k = V_s/V_p = 22/220 = 1/10 = 0.1$. Since $V_s/V_p = N_s/N_p \implies 1/10 = N_s/500 \implies N_s = 50$ turns. **Q45:** (A) Capacitance is the ratio of charge on a conductor to its potential ($C = Q/V$). SI unit: Farad (F). (B) Since the battery remains physically connected: (i) Potential difference remains exactly constant ($V = V_0$). (ii) Capacitance drastically increases K times ($C = KC_0$). (iii) Charge drawn from battery increases K times ($Q = CV = KC_0V_0 = KQ_0$). (iv) Electric field remains exactly constant ($E = V_0/d = E_0$). (v) Energy stored increases strictly K times ($U = \frac{1}{2}CV_0^2 = \frac{1}{2}(KC_0)V_0^2 = KU_0$). **Q46:** (A) Electromagnetic Induction (mechanical energy into electrical energy via changing flux). (B) Diagram showing Armature coil, Strong field magnets, Slip rings, Carbon brushes, External load. (C) Flux $\Phi = NBA\cos(\omega t)$. Induced emf $\varepsilon = -d\Phi/dt = -NBA\frac{d}{dt}\cos(\omega t) = NBA\omega\sin(\omega t)$. Thus $\varepsilon = \varepsilon_0 \sin(\omega t)$. (D) Draw a standard sine wave starting from origin, peaking at ε_0, crossing zero at $T/2$, negative peak at $3T/4$, and zero at T. **Q47:** (A) Draw a complete ray diagram with small objective and larger eyepiece. Object just beyond $F_o \implies$ real, inverted, magnified image A'B' within $F_e \implies$ final highly magnified, virtual image A''B'' at distance D. (B) $m = m_o \times m_e$. For objective, $m_o = v_o/u_o \approx L/f_o$. For eyepiece acting as a simple magnifier at D, $m_e = (1 + D/f_e)$. Total $m \approx -\frac{L}{f_o}(1 + \frac{D}{f_e})$. (C) Magnification $m \propto 1/f_o$ and $m \propto 1/f_e$. Extremely short focal lengths are explicitly required to produce a large magnifying power. **Q48:** (A) $d\vec{B} = \frac{\mu_0 I}{4\pi}\frac{d\vec{l} \times \vec{r}}{r^3}$. (B) Draw circular loop in y-z plane, observation point P on x-axis. $dB = \frac{\mu_0 I dl}{4\pi(R^2+x^2)}$. Resolve into axial ($dB\sin\theta$) and perpendicular ($dB\cos\theta$) components. Perpendicular strictly cancel. Net field $B = \int dB\sin\theta = \int \frac{\mu_0 I dl}{4\pi(R^2+x^2)}\frac{R}{\sqrt{R^2+x^2}} = \frac{\mu_0 IR}{4\pi(R^2+x^2)^{3/2}}(2\pi R) = \frac{\mu_0 IR^2}{2(R^2+x^2)^{3/2}}$. (C) For $x \gg R$, $B = \frac{\mu_0 IR^2}{2x^3}$. Multiply and divide by π: $B = \frac{\mu_0 I(\pi R^2)}{2\pi x^3} = \frac{\mu_0 (IA)}{2\pi x^3} = \frac{\mu_0}{4\pi}\frac{2M}{x^3}$, which is the precise field of a magnetic dipole on its axis. **Q49:** (A) Every point on a given wavefront explicitly acts as a fresh source of new secondary spherical wavelets spreading out at the speed of light. The forward envelope of these wavelets forms the new wavefront. (B) Draw incident plane wavefront AB at interface. Draw secondary wavelet from A in medium 2 with radius $v_2 t$. Draw tangent CD from B's refracted position. (C) In $\triangle ABC$, $\sin i = v_1 t/AC$. In $\triangle ADC$, $\sin r = v_2 t/AC$. Ratio $\frac{\sin i}{\sin r} = \frac{v_1 t/AC}{v_2 t/AC} = \frac{v_1}{v_2}$. By definition $\mu_1 = c/v_1, \mu_2 = c/v_2 \implies v_1/v_2 = \mu_2/\mu_1$. **Q50:** (A) $K_{max} = h\nu - \Phi$ (or $h\nu - h\nu_0$). K_{max} is maximum kinetic energy, h is Planck's constant, ν is incident frequency, Φ is strictly the work function. (B) Plot a straight line originating from the x-axis at ν_0 and sloping strictly upwards. (C) Comparing with $y = mx + c$, the straight-line slope mathematically equals Planck's constant ($m = h$). Extrapolating the line backwards, the negative y-intercept magnitude exactly yields the material's work function ($c = -\Phi$). (D) No, threshold frequency is solely an intrinsic property of the nature of the photosensitive material and is completely independent of the macroscopic intensity of the incident light.